Can't ship AI without knowing if it helps (or hurts!)

Importance of Metrics

Opening Scenario

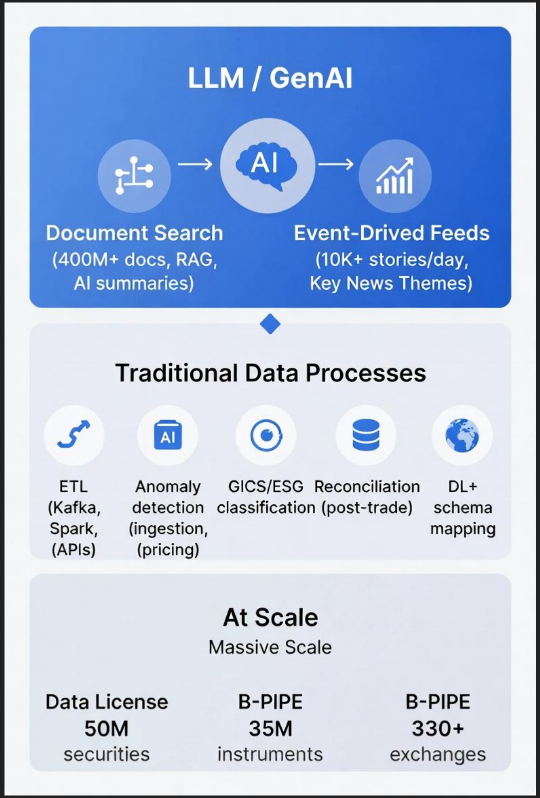
- A team ships an AI change that speeds up ingestion by 2×.
- Clients report **wrong data**.
- **What went wrong?**

We'll answer that today
We will show you how to **detect problems before clients do.**

What we will cover

1. Where AI Touches Bloomberg - Why it is important
2. Data Quality and its role - AI cannot fix cracks
3. Five Block Taxonomy
4. Metrics that matter (in practice)

Where AI Touches Bloomberg



Data Quality: AI can't fix cracks in the foundation

- **Every AI change must answer:** Did we maintain quality while improving speed or cost?
- **Data quality is the foundation:** AI can’t compensate for incorrect, missing, or stale inputs.
- **Operational wins must preserve usability:** Faster pipelines only help if outputs remain client-ready.
- **AI metrics aren’t enough on their own:** If accuracy or completeness drops, “better” model scores don’t matter.

Data Quality's Role

Data Quality Metrics

- **Accuracy:** Correct vs ground truth.
- **Completeness:** % missing in required fields, coverage per entity/time window.
- **Timeliness:** Freshness lag and update latency vs SLA.
- **Consistency:** Cross-source agreement rate, unit/label alignment.
- **Validity:** Schema + constraint pass rate (types, ranges, formats). Pydantic / Great Expectations rejects invalid rows early.

Five Block Taxonomy - Overall Evaluation of Workflows

- **Makes evaluation manageable:** One framework, five blocks, no guesswork on what to measure.
- **Prevents wasted optimisation:** Fix data and ops first, otherwise you'll "improve" AI on broken foundations.
- **Enables release discipline:** Metrics become gates and triggers, not just dashboard numbers.

Five Block Taxonomy: Questions and Decisions

Block	The question you're asking	Your decision	In one line
1. Data quality	Is the data itself correct enough to ship?	Ship / hold	"Can we trust the inputs?"
2. Operational workflow	Is the pipeline running reliably?	Deploy / rollback	"Does the system stay up?"
3. Human workflow	Did we reduce manual work without making things worse?	Automate / keep manual	"Did we save time safely?"
4. AI-specific	Did the AI step add value without introducing unreliability?	Promote / reject	"Is the AI helping or hurting?"
5. Governance & risk	Can we prove what happened and why?	Audit / remediate	"Can we explain and defend it?"

Five Block Taxonomy: When to use and Metrics

Block	When to use	Metrics	Bloomberg examples
1. Data quality	Before AI evaluation; fix data first	Accuracy, completeness, timeliness, consistency, validity	Data License; Event-Driven Feeds
2. Operational workflow	Before AI impact; pipeline must be stable	Lead time, change failure rate, MTTR	B-PIPE; Kafka pipelines
3. Human workflow	AI replaces/assists human tasks	Hours saved, error rate delta	Document Search
4. AI-specific	AI/ML step in path (RAG, classification, anomaly)	LLM: RAGAS, faithfulness, Pydantic; Traditional: precision, recall, F1	Event-Driven Feeds; NSTM; anomaly detection
5. Governance & risk	All AI products	Auditability, traceability, compliance	Document Search attribution; model/prompt versioning; audit records

Activity: Which Block is this Scenario?

Metrics: Three families + overarching

- Processing & operational
- GenAI & LLM
- Traditional ML
- Overarching: Budget, cost, data quality

Key Theme: Metrics preserve Trust - Wrong Data erodes Trust

Processing & Operational Metrics

- **Ingestion:** throughput, p95 latency, lag
- **Transformation:** accuracy, failure rate, automation coverage
- **Validation:** pass rate, schema conformance, freshness lag
- **Data management:** SLA compliance, storage cost, duplicate rate
- **Ops:** lead time, change failure rate, MTTR
- **Human:** hours saved, override rate, error delta

Gen AI & LLM Metrics

- **Key metrics:** faithfulness (groundedness), answer relevance, retrieval recall, retrieval precision, citation correctness, abstention rate
- **How we measure:** RAGAS + calibrated LLM-as-judge + targeted human review on a stratified sample
- **Guardrails:** Pydantic schema validation for outputs; MLflow (GenAI) for experiment tracking, eval runs
- **RAGAS (core):**
  - **Faithfulness (0–1):** compare to baseline; investigate drops or low scores
  - **Answer relevance:** does it answer the user question, not just paraphrase context
  - **Context recall:** did we retrieve the needed evidence
  - **Context precision:** how much retrieved context was actually useful (noise control)
- **Bloomberg examples:** Document Search; Event-Driven Feeds

Traditional ML Metrics

- For **classification**: Precision, recall, F1, time-to-detect;
- For **regression**: MAE, RMSE, R^2, directional accuracy
- Bloomberg examples:
  - **Anomaly detection**: Precision (few false alarms), recall, time-to-detect
  - **GICS/ESG classification**: Classification accuracy, inter-rater agreement
  - **Reconciliation & DL+ schema mapping**: Break detection rate, false break rate, time to resolve; mapping accuracy, time to first mapping, override rate
  - **Regression / forecasting**: Earnings surprise prediction (actual vs estimated EPS); MAE, RMSE, R^2, directional accuracy.

Overarching Metrics: Allowance, Cost

- **Error allowance**: Target 99.9% success → 0.1% can fail. Example: 1M requests = 1,000 failures allowed.
- **Cost metrics**: Cost per inference (e.g. $ per LLM call), compute cost (GPU/CPU), ROI of automation, infra + vendor spend
- **Resource utilisation**: CPU, memory, GPU usage; token limit. Relevant when AI scales in production.

AI that speeds up ingestion but doubles cost may not be a win.

Activity: Metric Game

Closing remark: The same Opening Scenario

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They shipped without measuring. They didn't gate on quality.
They hoped and didn't know.

The answer: Metrics preserve trust. Wrong data erodes it. Fix the foundation first.
Define success before you ship **(Five Taxonomy Blocks)**

Thank you for listening!